



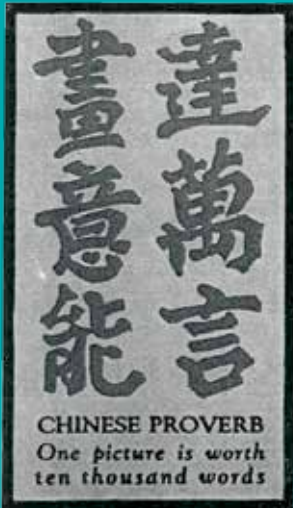
The Human Connectome Project

A compilation of neural data and tools to graphically navigate it, and further understand the human brain. <http://digit.in/mar21-13>



AI to detect Alzheimer's Disease

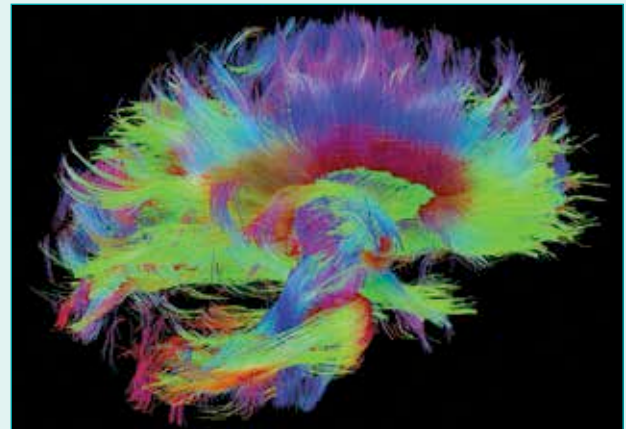
A new machine learning algorithm that detects signs of Alzheimer's disease six years before it is diagnosed. <http://digit.in/mar21-14>



Neuroimaging techniques

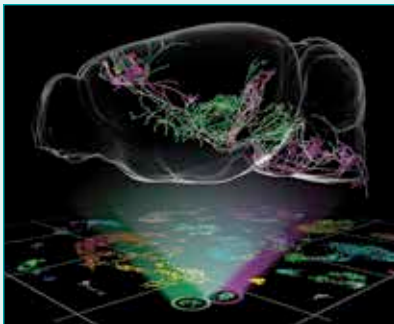
Novel neuroimaging, genetic and machine learning techniques are allowing researchers to peer into brains with unprecedented detail, prying apart its inner workings down to individual neurons.

Aditya Madanapalle | aditya@digit.in



Credit: Human Connectome Project

↑ The wiring diagram of the human brain produced by mapping the movement of water molecules using diffuse tensor magnetic resonance imaging. The research could one day be used to modify the connections in the brain to bring about the required behavioral changes.



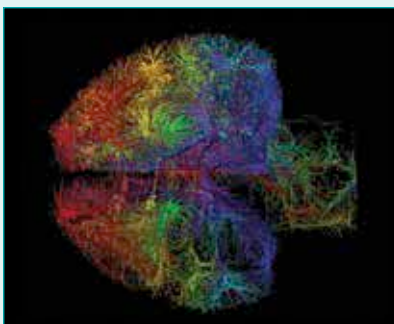
Credit: Allen Institute

← Two rare brain cells are highlighted in a mouse brain, after researchers mapped the gene expression in neurons and characterised them.



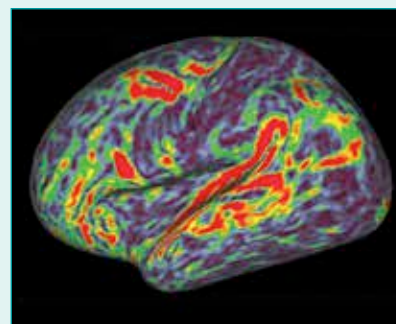
Credit: Anastasia Y, Viviana S/Harvard

← Maps like these use structural analysis to identify and predict progression of diseases, in this case, anxiety and depression among youth.



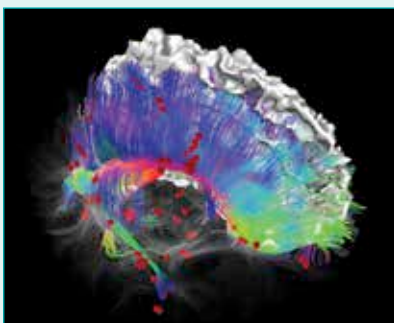
Credit: Allen Institute

← Another study modelling connections in a mouse brain identified long range connections that kept activities in different brain regions in sync.



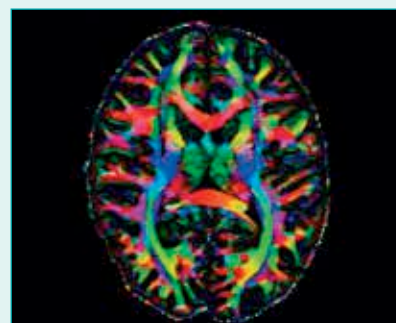
Credit: Matthew G, David VE, Washington University

← Blood flow MRI images allow researchers to investigate when the brain functions as a whole like liver tissue or when it has localised functional areas.



Credit: Interallia

← The brain of a patient with multiple sclerosis, the white matter tracts have damaged lesions, which are highlighted as the red boxes.



Credit: Human connectome project

← Shown here is an example of a colour fractional anisotropy map, where the colours represent the principal directions of the fibers in brain tissue.